
Jointly predicting RNase H-mediated gapmer potency and tolerability to reduce preclinical screening costs

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ABSTRACT

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1 Introduction

- Relative clinical success for 2-MOE's in CNS
- Despite this vast majority of amenable conditions remain undeveloped bc of cost constraints.
- ASOs require “thick” preclinical pipelines due to failure rates
-

2 Results

2.1 Fig 1: Dataset Overview

Message: “We assembled the largest public multi-endpoint ASO preclinical dataset, linking in vitro efficacy to in vivo toxicity for thousands of compounds.”

- **Panel A — Sankey diagram.** Data flow from in vitro inhibition → dose response → hepatic tox / neuro tox. Box heights $\propto$ measurements, flow widths $\propto$ compound overlap.
- **Panel B — Gene circle.** Donut chart of measurements per target gene aggregated across all four data types. Major genes shown individually; remaining grouped as “Other”.

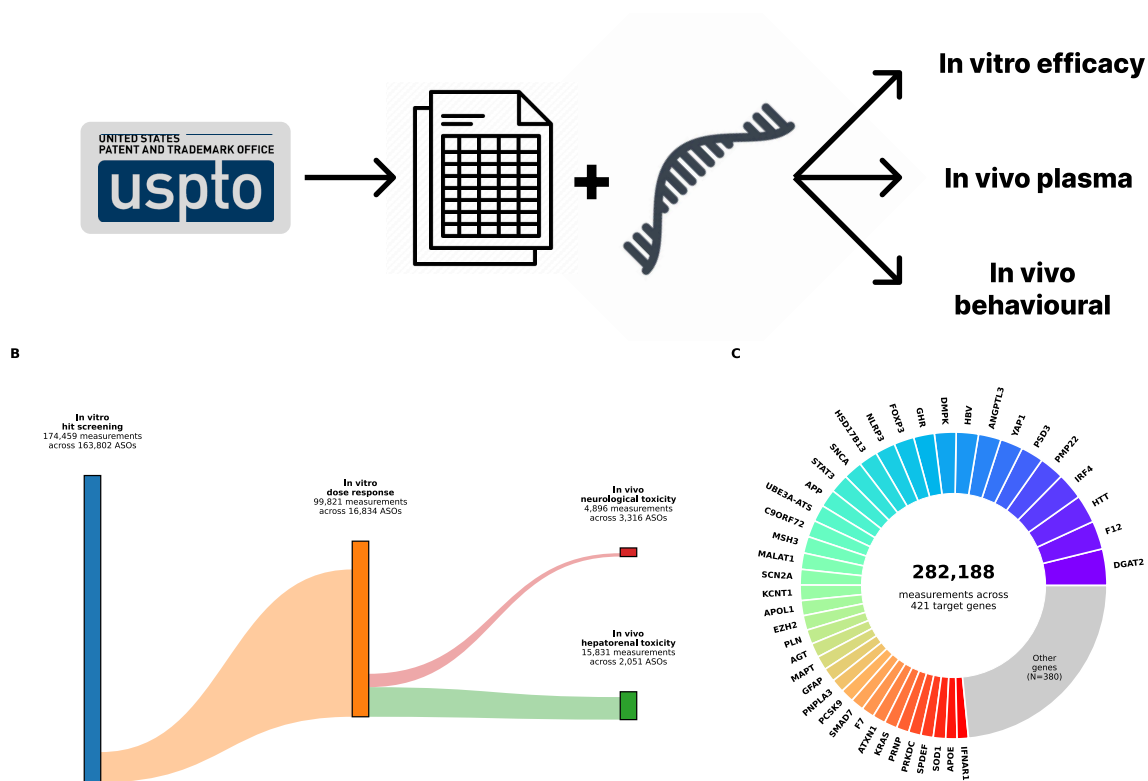


Figure 1: Overview of the OligoStack extraction pipeline and resulting dataset. **(A)** Three-stage pipeline converting USPTO patent tables into structured preclinical ASO datasets with HELM annotations. **(B)** Flow of compounds through in vitro hit screening, dose-response characterisation, and in vivo toxicity assessment; box heights are proportional to measurement counts, flow widths to the fraction of shared compounds. **(C)** Distribution of measurements across target genes; each coloured segment represents a major gene, with remaining genes grouped as “Other”.

2.2 Fig 2: The Pipeline Problem

Message: “Sequential screening requires ~2,000 initial candidates at ~\$3.5M to yield 3 development candidates. Most cost is concentrated at expensive in vivo stages where attrition is highest.”

- **Panel A — Pipeline attrition funnel.** 7 stages (inhibition >80%, IC₅₀ <500nM, mouse ALT <100, mouse FOB ≤1, rat ALT <100, rat FOB ≤1, monkey ALT <100). Shows ASO counts, pass rates, and per-stage costs.
- **Panel B — Stage measurement distributions.** 8-panel histograms with pass/fail thresholds (red dashed lines + green pass regions). Gives intuition for how stringent each cutoff is.
- **Status:** Both panels exist (pipeline_attrition.svg, pipeline_distributions.svg). Need composition into a single figure.

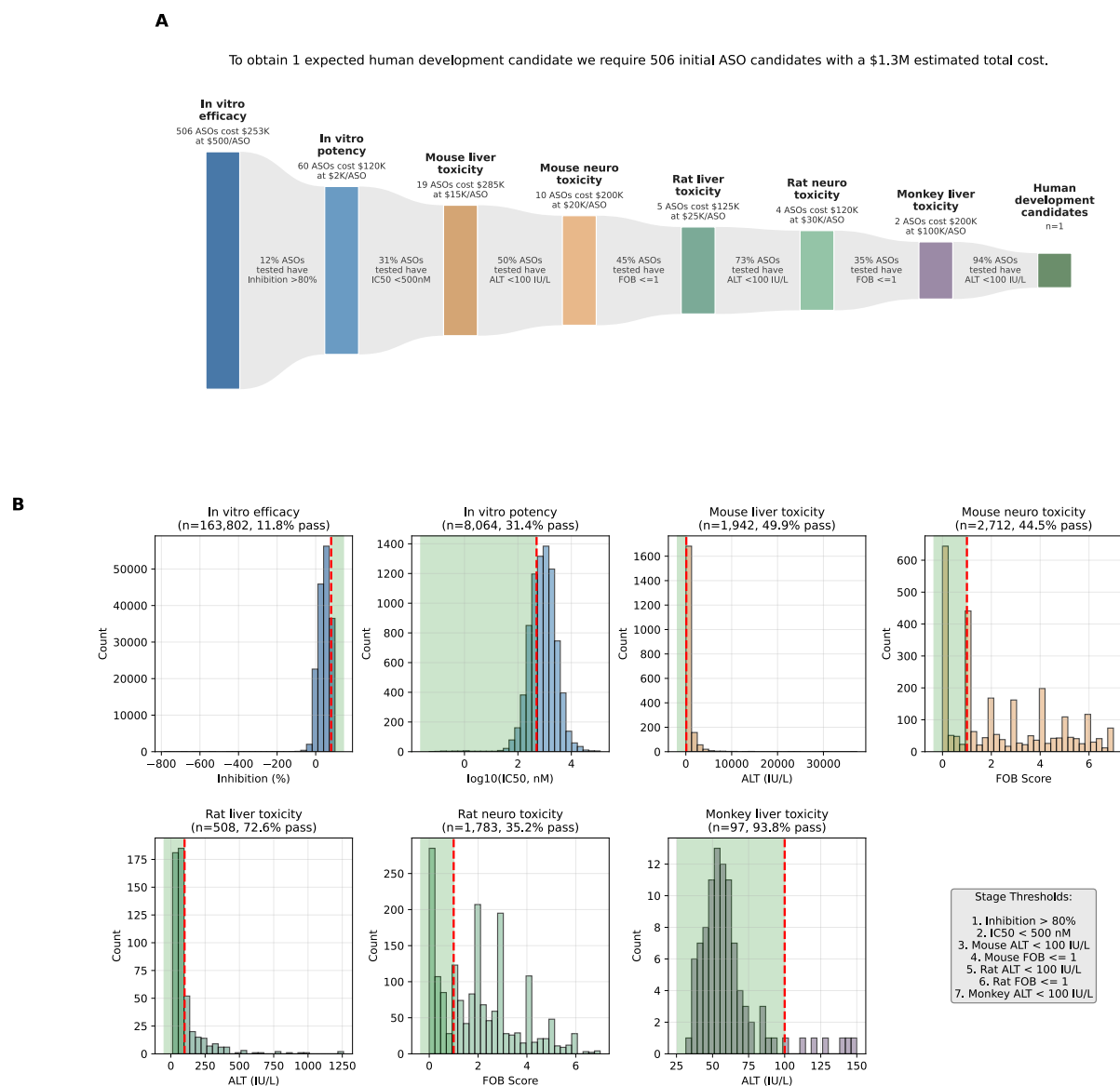


Figure 2: The ASO preclinical screening pipeline. **(A)** Attrition funnel showing the number of ASO candidates entering each stage, per-stage pass rates, and cumulative costs. Box heights are proportional to $\log(\text{ASO count})$. **(B)** Distribution of measurements at each pipeline stage with pass/fail thresholds (red dashed lines) and passing regions (green shading).

2.3 Fig 3: Model Overview

Message: “OligoAI2 jointly predicts ASO potency and tolerability from chemical structure alone, using a shared learned representation across in vitro and in vivo endpoints.”

- **Panel A — High-level architecture schematic.** Input: ASO chemical structure (HELM notation) + experimental covariates (dose, species, transfection). Middle: “Learned ASO representation” (black box). Output: 4 predicted endpoints (% inhibition, ALT, AST, FOB score). Annotate: trained on 249K in vitro + 27K in vivo measurements.
- **Panel B — Training strategy.** Simple 2-phase diagram: Phase 1 warms up on abundant in vitro data, Phase 2 jointly fine-tunes with scarce in vivo data. Emphasise the data imbalance problem this solves.

- **Status:** Does not exist. Best created as a clean vector schematic (Figma/draw.io). Keep deliberately simple — detailed architecture goes in Methods text.

2.4 Fig 4: Results and Pipeline Impact

Message: “OligoAI2 accurately predicts toxicity endpoints and, when used as a pre-screen, enriches the candidate pool 2–3× at toxicity stages — translating to significant cost savings.”

- **Panel A — Predicted vs. actual scatter plots.** One sub-panel per task (inhibition, ALT, AST, FOB). Density-coloured points, Pearson r annotated. Pass/fail thresholds marked as dashed lines (80% inhibition, 100 IU/L ALT, FOB ≤ 1). Test set, de-normalised to original units.
- **Panel B — Enrichment bar chart.** Grouped bars: base rate vs. top-10% pass rate for each endpoint, enrichment factor annotated (Inhibition 1.3×, ALT 2.1×, FOB 3.2×). Bridges from “model is accurate” to “model improves screening”.
- **Panel C — Cost comparison.** Stacked bar chart: Baseline vs. OligoAI2-guided pipeline costs, broken down by stage. Shows where savings concentrate (in vivo stages).
- **Status:** Panel C exists (pipeline_cost_comparison.svg). Panels A and B need creation via analyses/plotting/plot_figure4.py.

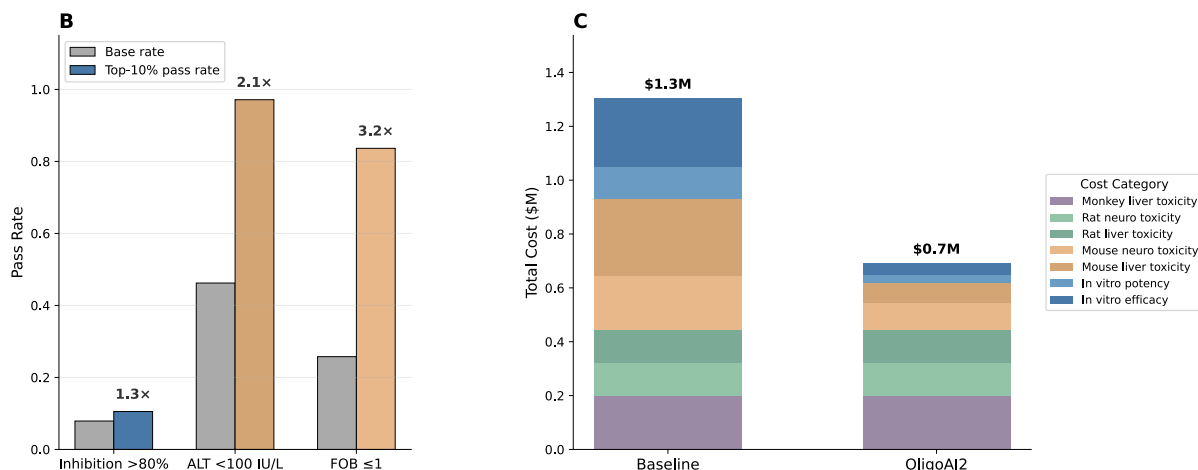


Figure 3: OligoAI2 enrichment and pipeline cost impact. **(B)** Grouped bars comparing the base pass rate (grey) with the top-10% model-selected pass rate (coloured) at each endpoint; enrichment factors annotated above. **(C)** Stacked bar chart of total pipeline costs by stage for the baseline screening pipeline versus the OligoAI2-guided pipeline.

3 Discussion

4 Methods

4.1 OligoStack Extraction Pipeline

OligoStack is a three-stage language-model-powered pipeline that converts unstructured tables in USPTO patent XML files into flat, structured preclinical ASO datasets annotated with HELM chemical-structure strings.

Stage 1 — Table extraction. USPTO patent XML documents filed after 2001 by ISIS Pharmaceuticals (now Ionis Pharmaceuticals) were retrieved via the USPTO bulk-data API and split into individual table files, each paired with the five paragraphs of prose immediately preceding the table. Tables were deduplicated in two phases: exact MD5 hash matching, followed

by pairwise sequence similarity (Python `SequenceMatcher`, 90% threshold) with length-based pruning; connected components were resolved by depth-first search to select a single canonical file per group, reducing 35,871 raw tables from 1,125 Ionis Pharmaceuticals patents to 8,435 canonical tables. For each canonical table, GPT-5-mini generated a bespoke Python extraction function tailored to that table’s XML layout. Generated scripts were executed in a sandboxed subprocess with a restricted import whitelist (`json`, `re`, `xml.etree.ElementTree`, `math`) and a ten-second timeout. An agentic self-repair loop allowed the model to observe execution output or errors and regenerate corrected code, for up to five attempts per table.

Stage 2 — HELM annotation. For each canonical table the model received the surrounding patent prose and a table preview, then generated a function that constructs Hierarchical Editing Language for Macromolecules (HELM) strings nucleotide-by-nucleotide from the chemistry description. To avoid hallucinated annotations, the function was required to return null when explicit modification data could not be found in the patent text. All HELM strings were validated by a rule-based checker that enforces correct sugar tokens (`[moe]`, `d`, `r`, `m`, `[cet]`, `[fR]`, `[lNa]`), canonical bases (`A`, `C`, `G`, `T`, `U`, `[5meC]`), backbone tokens (`[sp]`, `[am]`, `.`), balanced bracket syntax, and terminal-nucleotide constraints. **Stage 3 — Schema-driven collation.** Each assay category was extracted in a separate pass by supplying a schema dictionary that maps desired column names to natural-language descriptions. GPT-5 generated a mapping function per table that translates table-specific field names to the target schema, performs unit conversions, and unpivots multi-measurement rows into separate records. The same sandboxed execution and self-repair loop as Stage 1 were applied. HELM annotations were merged into each output record by compound identifier, following canonical-link chains across duplicate tables. Four schemas were defined, one per assay category: (i) *in vitro* inhibition — percent inhibition, cell line, dosage in nM, transfection method, treatment period, and target gene; (ii) dose response — the same fields with dosage in the original unit (nM or μ M); (iii) hepatorenal toxicity — seven serum biomarkers (ALB, ALT, AST, BUN, CREA, TBIL, protein/creatinine ratio), dosage, species, strain, number of doses, dosing period, measurement source, and administration route; and (iv) neurotoxicity — FOB score, dosage, species, strain, latency, administration method, and score type.

4.2 Data Collection and Preprocessing

All preclinical ASO data were extracted from USPTO patent filings using the OligoStack pipeline. Raw tables were parsed into four assay categories: *in vitro* hit screening (single-concentration percent inhibition), *in vitro* dose response (multi-dose percent inhibition), *in vivo* hepatorenal toxicity (serum biomarkers), and *in vivo* neurological toxicity (functional observational battery scores). Each category underwent a standardised cleaning pipeline described below.

Five HELM-level quality filters were applied uniformly across all four assay categories before any assay-specific cleaning. Compounds whose HELM annotation contained uncertain sugar or backbone tokens (indicated by `?` placeholders in the HELM string) were removed, as were sequences with ten or fewer nucleotides (likely extraction artefacts), uniformly modified compounds with no DNA gap (steric-blocking ASOs outside the scope of gapmer-focused analysis), homopolymer sequences comprising a single repeated nucleotide (extraction artefacts from poly-T or poly-A placeholder annotations), and naked DNA oligonucleotides lacking any sugar modifications (unmodified sequences where the extraction model could not identify the modification pattern).

In vitro inhibition. Inhibition values outside the range $[-1000\%, 100\%]$ were discarded as extraction artefacts. Cell line names and species labels were standardised to canonical forms (e.g. “A-431” $\rightarrow$ “A431”, “cynomolgus” $\rightarrow$ “monkey”). Duplicate measurements — defined as identical compound ID and inhibition value — were collapsed, retaining the most recent patent. This yielded 174,459 measurements across 163,802 ASOs.

Dose response. Dosages reported in μM were converted to nM. The same inhibition range filter and species standardisation were applied. Rows with non-positive dosages were removed. Cell line names were harmonised using a lookup table of 150+ known aliases. Deduplication on compound, dosage, and inhibition yielded 99,821 measurements across 16,834 ASOs.

Hepatorenal toxicity. Dosage strings were parsed into numeric values and units. Weekly dosing rates (mg/kg/wk) were converted to per-dose equivalents (mg/kg) using the reported number of doses and dosing period. Only subcutaneous and intraperitoneal administration routes with plasma or urine measurement sources were retained. Extreme dosages ($> 10,000$ mg/kg) were excluded. Biomarker values were range-filtered (ALT and AST: 10–50,000 IU/L; ALB: 1–100 g/dL). Rows sharing the same compound, species, dosing regimen, and administration method were collapsed, aggregating biomarker replicates into lists. Species and strain names were standardised. The final dataset contained 4,341 collapsed records across 2,051 ASOs, with 7 biomarker channels (ALB, ALT, AST, BUN, CREA, TBIL, protein/creatinine ratio).

Neurological toxicity. FOB (functional observational battery) score strings were parsed into numeric lists, retaining only values in the valid range $[0, 7]$. Species and strain names were standardised (e.g. “C57/B16 mice” $\rightarrow$ “C57BL/6 mice”). Since the raw neurotoxicity data lacked target gene annotations, compound-to-gene mappings were inferred in two steps: first by direct lookup against the in vitro and dose response datasets, then by patent-level majority vote for remaining compounds (assigning the most frequent target gene within each USPTO patent). This resolved target gene identity for 97% of neurotoxicity records. Deduplication on HELM annotation, species, administration method, score type, and dosage yielded 4,900 records across 3,316 ASOs.

Gene symbol mapping. Target RNA names from the patent text were mapped to canonical HGNC gene symbols using the Ensembl REST API, supplemented by a manually curated dictionary of 300+ aliases (e.g. “Tau” $\rightarrow$ *MAPT*, “PKK” $\rightarrow$ *PRKDC*, “K-Ras” $\rightarrow$ *KRAS*). After merging synonyms, the combined dataset spans 488 unique target genes and 280,027 total measurements.

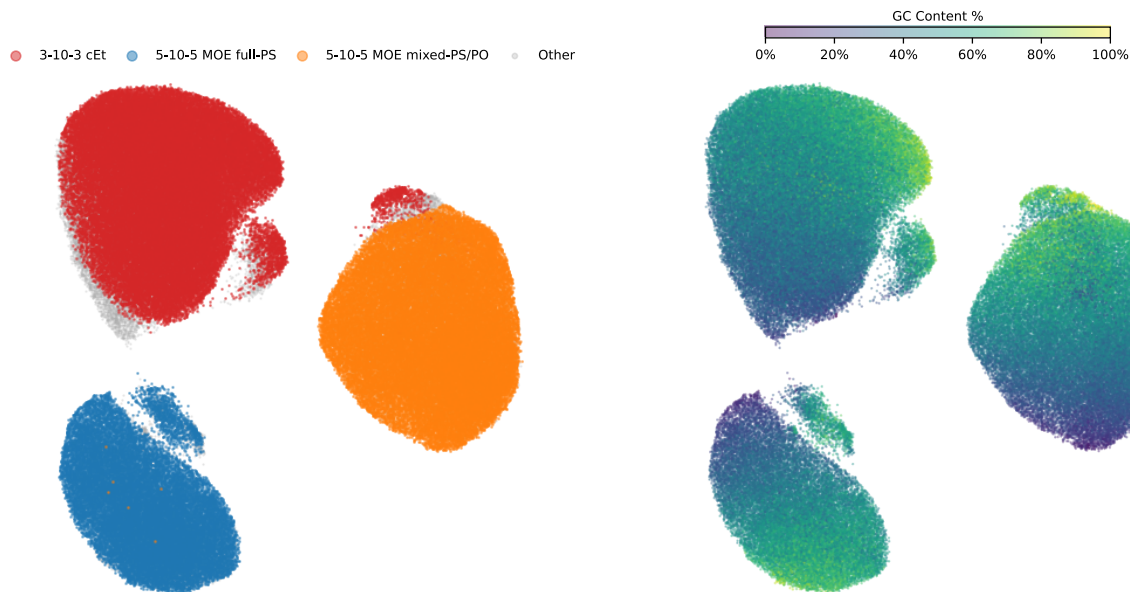


Figure 4: UMAP projection of OligoAI2 learned ASO representations, coloured by chemistry type (**left**) and GC content (**right**). Each point represents a unique HELM-annotated compound. The model’s bottleneck layer separates the two dominant gapmer designs — 5-10-5 MOE with full phosphorothioate backbone (blue) and mixed PS/PO backbone (orange) — as well as 3-10-3 cEt gapmers (red). Within each chemistry cluster, GC content varies smoothly, indicating that the embedding captures both chemical modification patterns and sequence composition.

4.3 Multi-Task Model

4.3.1 Problem Formulation

The preclinical ASO pipeline produces four heterogeneous endpoints: single-concentration percent inhibition, serum ALT, serum AST, and functional observational battery (FOB) score. We frame prediction as a four-task regression problem where each task shares a common learned ASO representation but retains independent output heads. The principal data challenge is a $\sim 30:1$ imbalance between the in vitro inhibition task (174,459 measurements) and the scarcest in vivo endpoint (FOB, 4,900 measurements). The model receives only the HELM chemical-structure string and experimental covariates (dose, species, transfection method) as input — no target-gene identity is provided, ensuring predictions generalise to novel gene programmes.

4.3.2 Model Architecture

The model ($\sim 304,504$ trainable parameters) consists of four modules:

- (i) **Factored HELM embedding.** Each nucleotide position is represented by the sum of three learned embedding vectors — one each for the base (A, C, G, T, U, 5-methylcytosine), sugar modification (MOE, DNA, LNA, cEt, fluoro-ribose, O-methyl, RNA), and backbone linkage (phosphorothioate, phosphodiester). This factored design captures the combinatorial chemistry of gapmer ASOs without an exponentially large vocabulary.
- (ii) **Transformer encoder with RoPE.** Two pre-norm transformer blocks with four attention heads, GELU-activated feed-forward layers, and rotary position embeddings

(RoPE) encode the nucleotide sequence. Attention masks ensure that padding positions do not contribute to the representation.

- (iii) **Attention pooling bottleneck.** A single learned query vector attends over the encoder’s output to produce a fixed-length ASO representation, replacing the more common masked mean pooling. This allows the model to weight nucleotide positions by their relevance.
- (iv) **Task-specific covariate encoders and heads.** Experimental covariates are encoded by separate small MLPs for in vitro (log-dose, treatment hours, transfection method) and in vivo (dose in mg/kg, number of doses, dosing period, administration route) contexts. Each task head is an independent two-layer MLP that receives the concatenation of the ASO representation and the covariate encoding.

4.3.3 Training Strategy

Training proceeds in two phases to address the 30:1 data imbalance. In Phase 1 (warmup), only the in vitro inhibition task is trained for 10 epochs, allowing the shared encoder to learn a meaningful ASO representation from the abundant in vitro data. In Phase 2 (joint), all four tasks are trained simultaneously with homoscedastic uncertainty weighting, where each task’s contribution to the total loss is scaled by a learned inverse-variance parameter σ_t^{-2} , following Kendall *et al.* (2018). This avoids manual loss-weight tuning and lets the model dynamically balance tasks of differing noise levels.

The per-task loss is the concordance correlation coefficient (CCC) loss, $\mathcal{L}_t = 1 - \text{CCC}_t$, which jointly penalises errors in correlation, mean bias, and variance mismatch. The total loss is $\mathcal{L} = \frac{1}{T} \sum_{t=1}^T \left[\frac{1}{2\sigma_t^2} \mathcal{L}_t + \log \sigma_t \right]$, where T is the number of active tasks.

Data are split by target gene to prevent information leakage: all measurements for a given gene appear exclusively in one of train (80%), validation (10%), or test (10%). This ensures the model is evaluated on its ability to generalise to unseen gene programmes. Optimisation uses AdamW with cosine annealing over 50 epochs and early stopping on validation loss (patience 10). A hyperparameter sweep over 14 configurations (learning rate, weight decay, dropout, encoder depth, warmup duration) selected the best model by median within-gene Spearman ρ on the validation set, achieving an overall median Spearman ρ of 0.445 across all four tasks.

To validate the contribution of in vitro data to in vivo prediction, we conducted an ablation study comparing three training conditions (Table 1): the full pipeline (warmup + joint training with IV data), joint training without warmup, and training on in vivo data only.

Table 1: Ablation study: effect of in vitro inhibition data on in vivo prediction. Median within-gene Spearman ρ on held-out test genes.

Condition	ALT ρ	AST ρ	FOB ρ	Median ρ
Full (warmup + joint)	0.352	0.249	0.632	0.352
No warmup (joint only)	0.465	0.428	0.519	0.465
Vivo-only	0.417	0.402	0.543	0.417

4.3.4 Evaluation

Model performance is assessed by two complementary metrics. **Ranking accuracy** is measured as the median Spearman rank correlation (ρ) computed within each target-gene group and aggregated across groups, rewarding the model’s ability to rank ASOs targeting the same gene

by their true endpoint values. Per-task Spearman ρ : in vitro inhibition 0.443, ALT 0.428, AST 0.446, FOB 0.708.

Top-K enrichment quantifies the model’s utility as a pipeline pre-screen. For each endpoint, compounds are ranked by model prediction and the top 10% are selected. The enrichment factor is the ratio of the pass rate among model-selected compounds to the base rate in the full test set: $EF = \frac{P(\text{pass} \mid \text{top-K})}{P(\text{pass})}$. An enrichment factor greater than one indicates the model concentrates passing compounds in its top predictions. We report enrichment at three pipeline stages: in vitro inhibition >80% ($EF = 1.3344\times$, $n=18,622$), ALT <100 IU/L ($EF = 2.1018\times$, $n=357$), and FOB ≤ 1 ($EF = 3.2453\times$, $n=551$).

5 Code Availability

Bibliography